

Samuel Cramer

Stakeholder	Wellbeing	Autonomy	Justice
Most Middlebury Residents	• Shorter routes to get to food establishments • Improved directions in terms of in-town transit • No option for other food establishments besides Two Brothers and Yogurt City • Many cause over-reliance on using the algorithm instead of just learning about the town's layout	• Free to use the algorithm • No personal data is stored or used • They know data is publicly available • No information about how the algorithm works • Forced to choose between two food establishments only	• Only works from 75 Shannon Street, preventing sensible use for those who don't use it as a frequent waypoint. • Disproportionately disadvantages those who do not like eating at those two food establishments • All users have access to the algorithm for free
Disabled Middlebury Residents / Wheelchair Users	• Could improve transit time for disabled residents when the route is accessible • Shortest path may not always be completely accessible • Disabled users can be misled into	• No knowledge of whether or not a route is accessible • Free to use the algorithm • No personal data stored or used • They know data is publicly available • No information	• Non-disabled users get full benefit, while disabled users may be harmed by being misled that a route is optimal when it is not accessible • Disabled residents who have

	thinking that it's also the best route for them when it is not	about how the algorithm works	advanced wheelchairs will be able to traverse all roads • Most disabled residents' wheelchairs will not be able to traverse an inaccessible route
Middlebury Restaurants	• Other restaurants may lose out on business • Restaurants further away from 75 Shannon Street are disadvantaged • May allow for more visibility of businesses on the way to TB or YC. • Two Brothers and Yogurt City have more pedestrian clients • Increases the value and profit of the target businesses	• No control of using algorithm to direct to these businesses • No way to opt out of having business in this algorithm • No information about how the algorithm works • No personal data is stored or used • The algorithm is free • They know data is publicly available	• Increased owner profits for TB and YC. • The benefits of increased foot traffic disproportionately go away from part-time workers • Disadvantage to businesses not selected in algorithm. • Less well-known restaurants have no easy path for being added to this algorithm.
Vermont Geodata	• Demonstrate faults in	• No control over use of	• VT Geodata

	dataset • No increased or decreased revenue • Data is being used for public benefit • Shows practicality of public data provisions	algorithm • VT Geodata doesn't know the algorithm exists • VT Geodata knows their data is publicly available	as a whole is credited for their dataset • The individuals who took the poor data for Franklin Street may be grilled by their employers for wrong classification • Lower-level worker's efforts won't be individually acknowledged
Town of Middlebury	• Increased walking in town may cause degradation of infrastructure or congestion • Increased driving time due to increased pedestrians • Roads ignored by the algorithm or dataset won't receive attention • Helps stimulate the local	• No control over use of algorithm • Town of Middlebury doesn't know the algorithm exists • The Town of Middlebury knows that VT Geodata's data is publicly available • Limited privacy concerns	• Residents living near common foot paths will see more traffic near them (pedestrian) • The algorithm uses visible roads equally. • Most residents aren't disadvantaged

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One area of ethical concern is the fact that the Bellman-Ford algorithm in this package only uses road edges and intersection vertices, but there is no particular inclusion of whether or not a road segment is accessible by a disabled individual. This means that a disabled user could be misled to traverse a treacherous path, which either makes travel dangerous, challenging, or even impossible, with no particular warning.

One way to ameliorate this is to incorporate Boolean accessibility flags as part of our analysis of road segments. In our ReadCSV.java file, we can see if the accessibility flag is actually met, and generate a Bellman-Ford graph corresponding to an accessible set of pathways and a non-accessible set of pathways. We'd only add an edge to the accessible set if the accessible flag is True. Then, in our Main.java file, we could ask our user if they would prefer an accessible path or not, and then use the rest of our package as is to have an accessible pathway between 75 Shannon Street and their restaurant of choice. This would improve safety for wheelchair users and make sure that they get the justice they deserve in the context of this algorithm.